

# DEPLOYMENT

## 1. Create directory and change directory

```
# mkdir 02.k8s-task
```

```
# cd 02.k8s-task
```

```
root@k8s-control:~# mkdir 02.k8s-task
root@k8s-control:~# cd 02.k8s-task/
root@k8s-control:~/02.k8s-task# pwd
/root/02.k8s-task
root@k8s-control:~/02.k8s-task#
```

## 2. Create manifest file

```
# vim game-2048.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: game
  labels:
    app: game
spec:
  replicas: 1
  selector:
    matchLabels:
      app: game
  template:
    metadata:
      labels:
        app: game
    spec:
      containers:
        - name: game
          image: marcincuber/2048-game
          ports:
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: svc-2048
spec:
  type: ClusterIP
  selector:
    app: game
  ports:
    - port: 8080
      targetPort: 80
      protocol: TCP
```

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: game
  labels:
    app: game
spec:
  replicas: 1
  selector:
    matchLabels:
      app: game
  template:
    metadata:
      labels:
        app: game
    spec:
      containers:
        - name: game
          image: marcincuber/2048-game
          ports:
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: svc-2048
spec:
  type: ClusterIP
  selector:
    app: game
  ports:
    - port: 8080
      targetPort: 80
      protocol: TCP

```

### 3. Apply file

```

# kubectl apply -f game-2048.yaml
# kubectl get pods -o wide
# kubectl get svc -o wide
# kubectl get deployments.apps game -o wide

```

```

root@k8s-control:~/02.k8s-task# kubectl apply -f game-2048.yaml
deployment.apps/game created
service/svc-2048 created
root@k8s-control:~/02.k8s-task# kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE             NOMINATED NODE   READINESS GATES
game-59fd974f46-vzfn8              1/1     Running   0           14s   192.168.69.196  k8s-worker02     <none>            <none>
root@k8s-control:~/02.k8s-task# kubectl get svc -o wide
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE   SELECTOR
kubernetes ClusterIP   10.96.0.1     <none>         443/TCP    10d   <none>
svc-2048  ClusterIP   10.96.198.172 <none>         8080/TCP   21s   app=game
root@k8s-control:~/02.k8s-task# kubectl get deployments.apps game -o wide
NAME      READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES               SELECTOR
game      1/1     1             1           30s   game         marcincuber/2048-game app=game
root@k8s-control:~/02.k8s-task#

```

### 4. Port forwarding

```

# kubectl port-forward svc/svc-2048 8080:8080 --address=0.0.0.0

```

```

root@k8s-control:~/02.k8s-task# kubectl port-forward svc/svc-2048 8080:8080 --address=0.0.0.0
Forwarding from 0.0.0.0:8080 -> 80
Handling connection for 8080
Handling connection for 8080

```

5. Go to web and check <http://192.168.153.10:8080>

